Neuron mechanisms II: what $d8$-dominant neurons compute

`wd3_k0_N2048`, step 500k — post-nonlinearity units, uniform evaluation data

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Setup

We use the same protocol as in the $d4$ report: $a = \phi(z)$, $\phi = \text{relu}(\tanh)$, $z = (Vh)_n$, analyzed on all 2^{16} uniform inputs. A neuron is called $d8$-*dominant* when the largest Walsh coefficient among the tracked tree modes is either χ_{d8_0} (bits 0–7) or χ_{d8_1} (bits 8–15). There are 2047 such neurons (307/452/619/669 in blocks 0–3). For each neuron, we decompose the pre-activation z in two complementary ways: over the full 256-mode Walsh basis of the neuron's own 8-bit half-cube, and over the three tree-level modes $y_0 = \chi_{d8_0}$, $y_1 = \chi_{d8_1}$, and $\chi_{d16} = y_0 y_1$. Looking at where the variance of z is concentrated cleanly separates the neurons into three mechanisms.

Three mechanisms, ordered by depth

(1) Blocks 0–1: 8-input majority gates on raw bits (951 neurons; 100% of block 0, 94% of block 1). Their pre-activation is well approximated by $z \approx \gamma \sum_{k=1}^{8} s_k x_k$, with $\gamma \approx 0.13$, and at least 99% of $\text{Var}(z)$ lies in the eight degree-1 singleton modes. This is the same mechanism as for the $d4$ neurons, but now applied to eight raw bits instead of four. The measured response as a function of the signed input sum, $E[a \mid s]$, matches the ideal $\text{relu}(\tanh(\gamma s))$ gate essentially exactly (Fig. 1, left). Because such a gate has Walsh power spread over many modes of the 8-bit cube, only a small fraction of its variance lies in the tracked binary-tree modes: the ideal coverage is 0.042, and the measured median is 0.044. This continues the same trend seen at lower degree: $d2$ corner units have coverage 1/3, $d4$ gates have coverage ≈ 0.15, and $d8$ gates have coverage ≈ 0.044. Figure 2 shows the full 255-mode Walsh spectrum such a gate produces, in direct analogy to the $d4$ spectrum figure: the eight singletons carry $\approx 9\%$ each, all 28 pairs are equal at $\approx 0.8\%$ (the four tree pairs are in no way special), odd degrees are absent, and the $d8$ mode itself is a single $\approx 1.1\%$ spike.

(2) Blocks 2–3: 8+8→16 corner units (189 neurons). Their pre-activation is a nearly perfectly balanced sum of the two already-formed $d8$ parities, $z \approx \gamma(\pm y_0 \pm y_1)$, with the variance split 49.7%/49.8% between the two inputs and essentially no $d16$ component already present in z. After the nonlinearity, these neurons become the same two-input corner detectors discussed in the $d2$ case: $a = \frac{t}{4}(1 \pm y_0 \pm y_1 \pm y_0 y_1)$. In other words, the neuron responds strongly to one particular sign combination of (y_0, y_1) (for example, 0.42 on $(+,+)$ and ≤ 0.005 on the other three corners; Fig. 3). Its variance then splits into **approximately equal thirds over** $\{\chi_{d8_0}, \chi_{d8_1}, \chi_{d16}\}$ (0.332/0.333/0.319 for the exemplar). *These neurons are therefore the point where the degree-16 target is actually created.* They are labeled "$d8$-dominant" only because the three coefficients are almost tied and small sampling differences make one of the $d8$ modes slightly larger. Since all three modes belong to the tracked tree, their total tree coverage is ≈ 1.0.

(3) Blocks 2–3: amplifiers (296 neurons). For these neurons, at least 90% of the pre-activation variance already lies in the neuron's own $d8$ feature assembled upstream, so the neuron mainly strengthens or passes along an existing $d8$ representation rather than constructing it from scratch. Their median tree coverage is 0.98. The remaining "other" class (600 neurons, median coverage 0.93) forms a continuum between the clean amplifier and corner mechanisms: these neurons still carry substantial $d8/d16$ structure, but with a larger contribution from additional modes, plus a small noisy tail. In fact, much of the "other" class is *effectively dead*: 487 of the 2047 $d8$-dominant neurons have total activation Walsh power $\text{Var}(a) < 10^{-4}$ (28%/45% of blocks 2–3), and 380 of them are "other". Figure 4 repeats the composition of Fig. 1 (right) with this taken into account: once neurons are weighted by their amplitude $\text{Var}(a)$, the picture sharpens — blocks 0–2 are dominated by raw-bit gates, and block 3 by the 8+8 corner units (64%), whose class total is carried largely by a few dozen strong units.

Takeaway. The network does *not* appear to build $d8$ by composing already-formed $d4$ features. Instead, blocks 0–1 create $d8$ directly from raw degree-1 inputs using wide 8-input majority gates. This is the same basic mechanism seen for $d4$, but because the gate acts on more inputs, an even smaller fraction of its Walsh spectrum

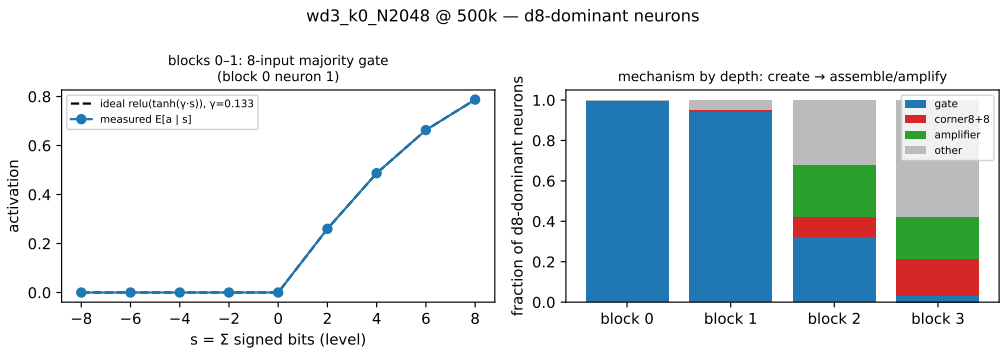


Figure 1: Left: a block-0 $d8$ gate's measured response as a function of signed input sum, compared with the ideal relu(tanh(γs)) curve (dashed; the two are essentially indistinguishable). Right: mechanism composition by depth — raw-bit gates create $d8$ early, while corner units and amplifiers become dominant in blocks 2–3.

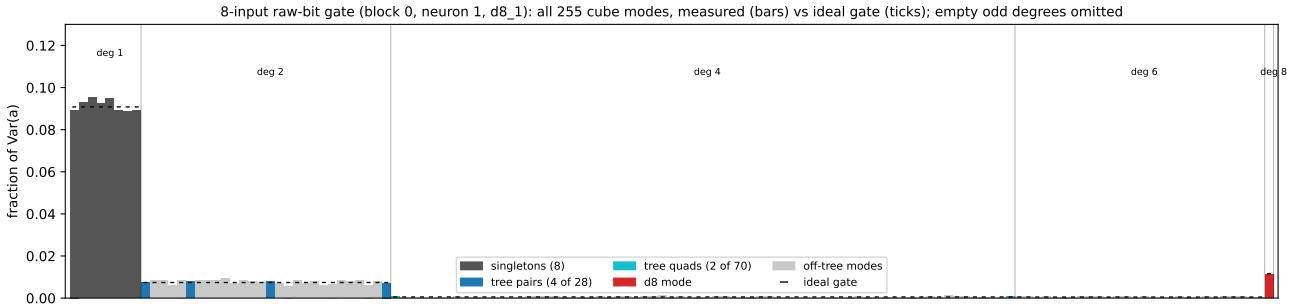


Figure 2: Activation Walsh spectrum of an 8-input raw-bit gate (block 0, neuron 1) over all 255 modes of its 8-bit cube, measured (bars) vs. the parameter-free ideal gate (ticks), grouped by degree (the empty odd degrees are omitted). Colors mark the tree-tracked modes; the tree pairs and quads sit at exactly the same height as their off-tree siblings, which is why tree coverage is only ≈ 0.044.

lies in the tracked tree (≈ 0.04). The only clear compositional step in this network occurs at the final level: dedicated corner units take the two assembled $d8$ parities as inputs and create χ_{d16} in blocks 2–3. This agrees with the layer-level decode, where the $8{+}8\,|\,16$ partition becomes dominant at exactly those blocks. Thus the same basic gate/corner mechanism appears across degrees; what changes is which features feed into it (raw bits at lower levels versus already-assembled parities at the final level) and how much of the resulting Walsh spectrum falls inside the binary-tree modes we track.

Artifacts: `gdrive:parity/analysis_output/neuron_mechanisms/wd3_k0_N2048/step_00500000/` (`d8_classes.csv`, `d8_per_neuron.csv`, spectra, figures, this report; companion $d4$ and $d2$ probes in the same folder).

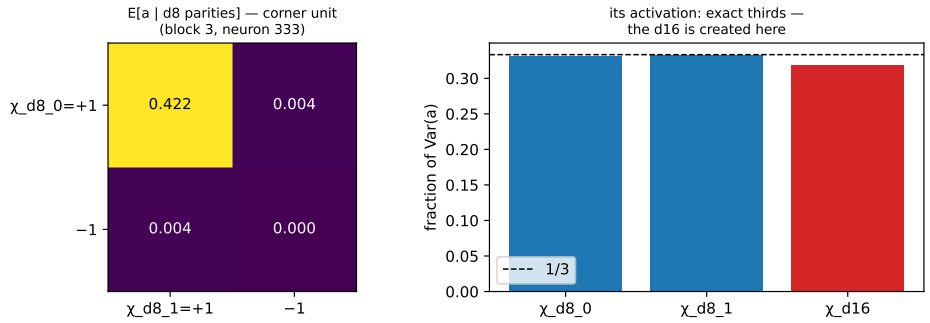


Figure 3: An 8+8→16 corner unit (block 3, #333). Left: the neuron fires almost only for one of the four possible sign combinations of the two $d8$ parities. Right: its variance splits into approximately equal thirds across the two $d8$ modes and the $d16$ mode — the $d16$ component is created by this neuron.

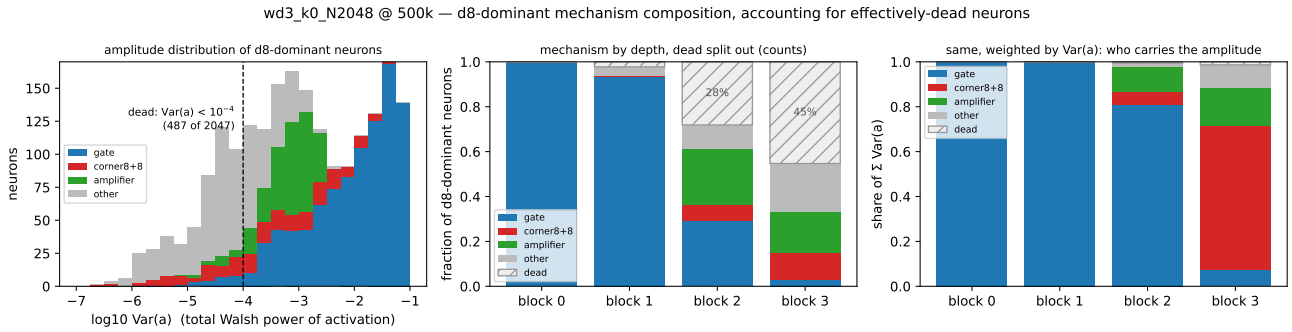


Figure 4: The composition of Fig. 1 (right), accounting for effectively-dead neurons ($\mathrm{Var}(a) < 10^{-4}$). Left: amplitude distribution by class — 487 of the 2047 $d8$-dominant neurons are dead, mostly "other". Middle: the same counts with the dead split out (28%/45% of blocks 2–3). Right: composition weighted by $\mathrm{Var}(a)$: the block-3 amplitude is carried mainly by the 8+8 corner units (64%, dominated by a heavy tail — the top 10 corners hold half the class total), while the dead sliver carries $\leq 1.3\%$ everywhere.

class	input (z)	blocks	n	tree coverage (median)
majority gate	$\gamma \sum_8 s_k x_k$ (raw bits)	0–1 (some 2)	951	0.044
8+8 corner	$\gamma(\pm y_0 \pm y_1)$	2–3	189	1.00
amplifier	own χ_{d8} feature	2–3	296	0.98
other (mixtures)	partial $d8/d16$ + bulk	2–3	600	0.93